PANS ACADEMIC REPOSITORY - FACULTY OF PHARMACY

DEPARTMENT OF PHARMACEUTICAL CHEMISTRY

Course: PCH 401 - SAR & Rational Drug Design

Session: 2024/2025 First Semester Degree Examination

TIME ALLOWED: 3 HOURS | INSTRUCTIONS: Answer ALL questions in Section A.

QUESTION 1 [Compulsory - 25 Marks]

(a) Define Structure-Activity Relationship (SAR) and define the term pharmacophore.

(b) Draw and label the Penam core nucleus of beta-lactam antibiotics.

(c) Describe how steric hindrance on the acylamino side-chain confers beta-lactamase resistance.

QUESTION 2 [15 Marks]

(a) Differentiate between classical and non-classical bioisosterism with two drug examples.

(b) Discuss the relationship between log P (partition coefficient) and membrane permeability.